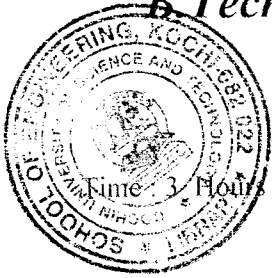


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B. Tech. Degree V Semester Examination November 2018



ME 15 - 1504 THERMAL ENGINEERING (2015 Scheme)

Maximum Marks : 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Compare solid, liquid and gaseous fuels.
- (b) Explain with a neat sketch the working of a Bomb calorimeter.
- (c) Derive an expression for the efficiency of an Otto cycle in terms of compression ratio ' r ' and ratio of specific heats ' γ '.
- (d) Explain actual valve timing diagram of a four stroke Petrol engine.
- (e) Explain Morse test on an IC engine.
- (f) What is supercharging? What are its objectives?
- (g) Explain with a neat sketch, the working of a diesel fuel pump.
- (h) Explain Wankel engine with a neat sketch.
- (i) Explain Pre-ignition and its effects.
- (j) Explain Octane number and Cetane number.

PART B

(4 × 10 = 40)

- II. (a) Explain adiabatic flame temperature. (3)
 - (b) The following is the ultimate analysis of a sample of petrol by weight: (7)
Carbon = 85%, Hydrogen = 15%. Calculate the ratio of air to petrol consumption by weight, if the volumetric analysis of the dry exhaust gas is : $\text{CO}_2 = 11.5\%$; $\text{CO} = 1.2\%$; $\text{O}_2 = 0.9\%$; $\text{N}_2 = 86\%$. Also find the percentage excess air.
- OR**
- III. (a) Explain the working of Junkers gas calorimeter. (3)
 - (b) Calculate the higher and lower heating values at constant pressure per kg of mixture at 25°C , for the stoichiometric mixtures of : (i) Air and Benzene vapour (C_6H_6) (7)
(ii) Air and Octane vapour (C_8H_{18}). The enthalpies of combustion at 25°C are $\text{C}_6\text{H}_6 = -3169500$ kJ/mole, $\text{C}_8\text{H}_{18} = -5116200$ kJ/mole, when water in the products are in the vapour phase.
- IV. (a) Obtain the expression for air standard efficiency of a diesel cycle in terms of compression ratio, cut-off ratio and ratio of specific heats. (2)
 - (b) An engine with 200 mm cylinder diameter and 300 mm stroke works on theoretical Diesel cycle. The initial pressure and temperature of air used are 1 bar and 27°C . The cutoff is 8% of the stroke. Determine : (i) pressure and temperature at all salient points (ii) Theoretical air standard efficiency (iii) Mean effective pressure (iv) Power of the engine if working cycles per minute are 380. Assume that compression ratio is 15 and working fluid is air. Consider all conditions to be ideal. (8)

OR

(P.T.O.)

- V. (a) What is meant by scavenging in IC engines? Explain. (2)
- (b) The following observations were recorded in a test of 1 hour duration on a single cylinder oil engine working on four stroke cycle. Bore = 300 mm, Stroke = 450 mm, Fuel used = 8.8 kg, Calorific value of fuel = 41800 kJ/kg, Average speed = 200 rpm. Mep = 5.8 bar, Brake friction load = 1860 N, Quantity of cooling water = 650 kg, Temperature rise = 22°C, Diameter of the wheel = 1.22 m. Calculate : (i) mechanical efficiency (ii) brake thermal efficiency. Also draw the heat balance sheet. (8)
- VI. Explain with neat sketch: (3)
- (a) Carburetor with compensating jet (3)
- (b) Magneto ignition system (4)
- (c) Full pressure lubrication system. (4)
- OR**
- VII. Explain with neat sketch: (3)
- (a) Free piston engine. (4)
- (b) Stratified charged engine. (3)
- (c) Clutch mechanism of an automobile. (5)
- VIII. Explain : (5)
- (a) Effects and control of detonation. (5)
- (b) Induction swirl and induction ports in SI engines. (5)
- OR**
- IX. Explain : (5)
- (a) Combustion phenomenon in CI engines. (5)
- (b) Types of combustion chambers in CI engines. (5)

